



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2023 – 0060



Agricultural Machinery – Front Loader HANIL HL25

Test Report Number	2023 – 0060
Date of Issuance	March 3, 2023
Valid Until	March 3, 2028
Agricultural Machinery	Front Loader
Type	Bucket-type, Tractor-mounted, Tractor remote hydraulics-powered
Brand	HANIL
Model	HL25
Test Requested by	FIT COREA TRADING PHILS., INC. 10 D. Arellano Street, Brgy. 135, Caloocan City, Metro Manila

SUMMARY

Performance Criteria	Manufacturer's Specification	AMTEC Test
Loading Capacity, kg/load	ND ^a	497.55
Loading Rate, kg/h	ND ^a	65119.80
Maximum Lift Height, mm	2679 ^b	2750

^a Not specified by the requesting party.

^b The data were obtained from the specification sheet provided by the requesting party.

Note: ND – No Data

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Provided
e. Maintenance instruction	Provided
f. Storage	Not provided
g. Handling, reception, transportation, assembly, and installation	Provided
h. Specifications	Provided
i. Dismantling and disposal	Dismantling only
j. Warranty	Provided
k. Parts list	Provided
2. Previous test report of similar brand and model	None
3. Manufacturer	HANIL INDUSTRY CO., LTD. Gyeongsan-si, Gyeongsangbuk-do, Korea
4. Four-wheel tractor used	40.5 kW TYM 5825R
5. Serial number	HL 22R015
6. Number of operators	One
7. Mounting/dismounting	Tools are needed to mount or dismount the front loader to or from the tractor.
8. Transporting the machine	The front loader is lifted by the four-wheel tractor during transport using lower link hydraulics.
9. Adjusting and repair of parts	The front loader is accessible for repairs. Adjustment of parts was not verified.
10. Loading and unloading of soil	The operator controlled the front loader through its remote hydraulic system. The soil was scooped from the ground using the bucket and then lifted during transport going to the truck. The bucket is then shifted to dumping position for unloading soil to the truck.

OBSERVATIONS (Cont'n)

11. Failures or abnormalities that may be observed on the implement or its component parts during and after the operation	No failure or abnormalities were observed on the machine or its component parts during and after the test operation.
12. Presence of manufacturing defects	No manufacturing defects were observed on the machine or its component parts during the test operation.
13. Rust-free and properly painted metal surfaces	The machine is rust-free and properly painted.
14. Presence of sharp edges and rough surfaces	No sharp edges or rough surfaces were observed on the machine except for the ground-engaging tool during the test operation.
15. Provision for the operator's safety	Warning notices are provided on the front loader.
16. Other remarks	A backhoe was attached at the rear of the four-wheel tractor.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	February 9, 2023
Location of Test	Brgy. Unang Sigaw, Quezon City 14°39'39.6" N, 120°59'56.4" E
Items Tested	Operator's Manual, Loading Capacity, Maximum Lift Height, Fuel Consumption Rate, Noise Level, and Number of Operators

METHODS OF TEST

The front loader was tested based on the AMTEC Methods of Test.

DESCRIPTION OF THE MACHINE

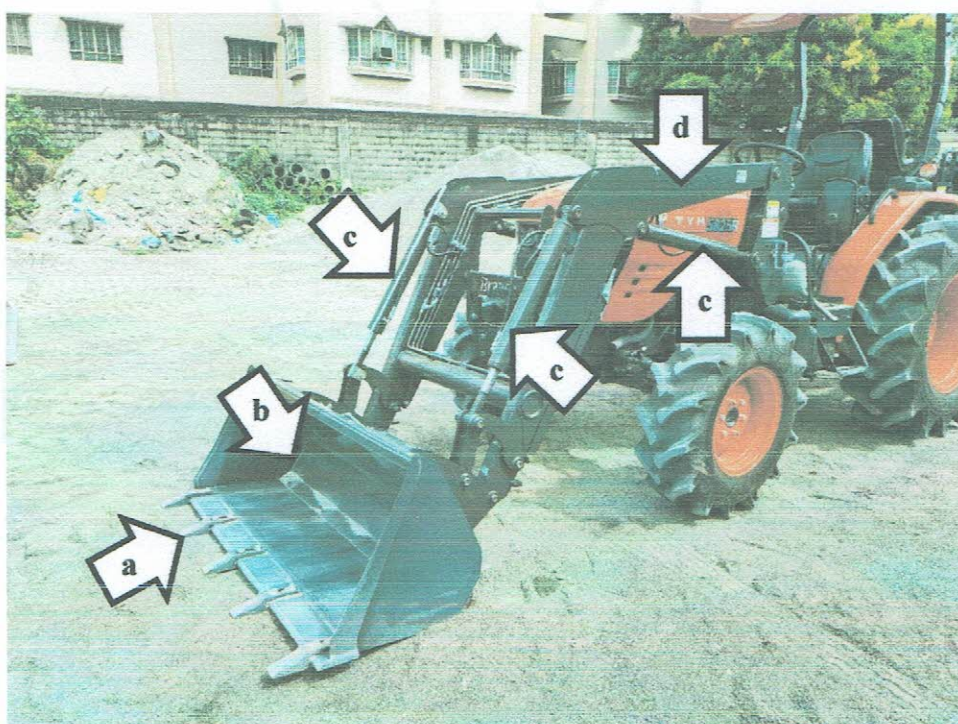


Figure 1. Parts of the HANIL HL25 Front Loader mounted to 40.5 kW TYM 5825R Four-Wheel Tractor include the following: (a) Ground-engaging tool, (b) Bucket, (c) Hydraulic cylinders, and (d) Arm.

SPECIFICATIONS

Item	Manufacturer's Specifications ^a	AMTEC Verification
1 Overall dimensions (transport mode) and weight		
1.1 Length, mm	800	2560
1.2 Width, mm	1050	1500
1.3 Height, mm	2200	1534
1.4 Weight, kg	429	NM ^b
2 Total arm length, mm	1900	2240
3 Bucket		
3.1 Dimensions, L × W × H, mm	1500 × 570 × 615	1510 × 695 × 485
3.2 Opening dimension, L × W, mm	ND	1455 × 525
3.3 Shape	Straight (Non-curved)	Triangular
3.4 Capacity, m ³	0.32	0.369 ^c
4 Maximum lift height, mm	2679	2750
5 Reach, mm		
5.1 Maximum reach of bucket at maximum lift height	251	1270
5.2 Maximum reach of bucket	ND	1755
5.2.1 Maximum reach of bucket at collecting position on the ground	1799	1766
5.2.2 Bucket angle, degree		
5.2.2.1 Maximum dump angle, degree	57	103 ^d
5.3 Maximum rollback angle, degree	42	58 ^d
6 Ground clearance at transportation mode, mm	ND	660

^a The data were obtained from the operator's manual provided by the requesting party.

^b Weight was not verified as there was no available scale on site.

^c Measured based on loading capacity during the test.

^d Measured from the horizontal axis.

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
7	Tractor used		
7.1	Brand	TYM	TYM
7.2	Model	5825R	5825R
7.3	Serial number	CNBB00002	ND
7.4	Make	Korea	Korea
7.5	Rated power, kW	40.5	40.5 ^a
7.6	Rated speed, rpm	2600	2600 ^a
7.7	Displacement, cm ³	2286	2286 ^a
7.8	Fuel type	Diesel	Diesel
7.9	Cooling system	Water-cooled	Water-cooled
7.10	Starting system	ND	Electric
8	Safety devices	ND	None
9	Special features	ND	None

^a The data were obtained from the operator's manual and nameplate of the tractor provided by the requesting party.

Note: ND – No Data, NM – Not Measured

RESULTS

Item		Data
1	Soil conditions	
1.1	Soil type	Sand
1.2	Bulk density, kg/m ³	1347.12
1.3	Moisture content, % _{d.b.}	3.93
2	Site condition	
2.1	Location	Brgy. Unang Sigaw, Quezon City
3	Field performance test	
3.1	Tractor gear setting	1 st gear - Medium
3.2	Loading capacity, kg/load	497.55
3.3	Loading rate, kg/h	65119.80
3.4	Fuel consumption rate, L/h	4.25
3.5	Noise at operator's ear level, dB(A)	89.7

OBSERVATIONS



Figure 2. Front loader during operation.



Figure 3. Nameplate of the HANIL HL25 Front Loader.

OBSERVATIONS (Cont'n)



Figure 4. Marking and warning signs on the HANIL HL25 Front Loader.

Tested by:


KAREN CYBILL F. CABANGLAN

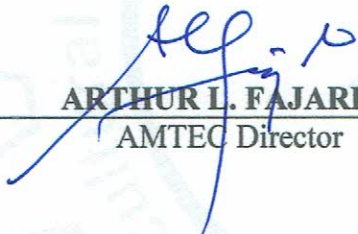
Test Engineer

Verified by:


ALPHONSE REGIN R. MAGMANLAC

Test Engineer

Approved for Release:


ARTHUR L. FAJARDO

AMTEC Director

MAR 06 2023

Date

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